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Mathematical Preliminaries and Error Analysis

Chapter Learning Outcomes

CLO

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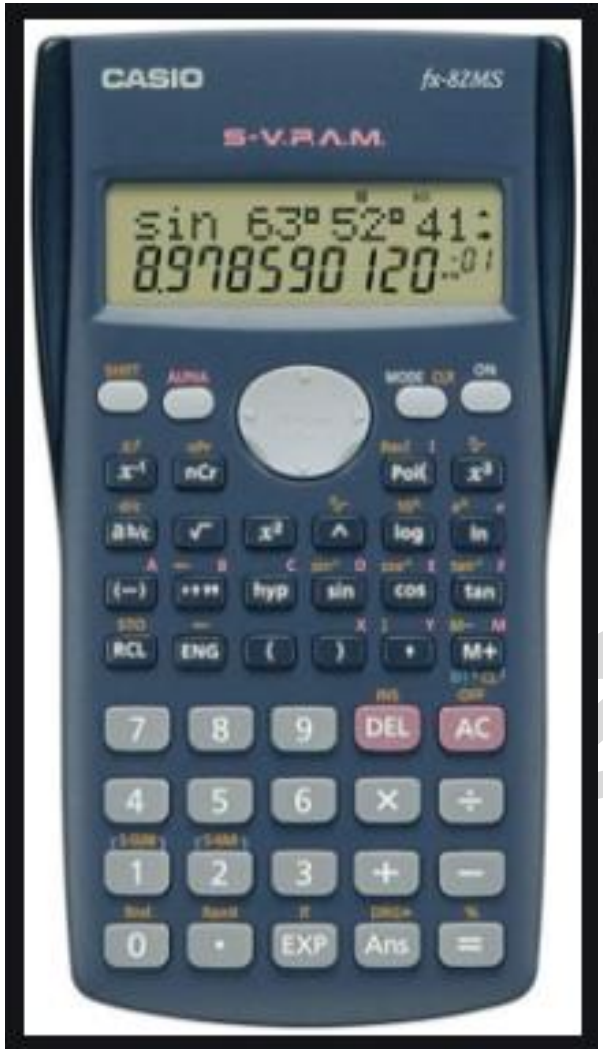
- Define the numerical analysis and determine when do we need numerical methods?
- Represent a real number in a normalized decimal form.
- Compute the absolute, relative, and relative percentage errors.
- Distinguish between rounding and chopping.
- Solve some problems using Finite-Digit Arithmetic.

You should have a scientific calculator



Please, put your calculator in radians mode !

Try to find sin 30 >>>>

0.5	Degree Mode
-0.988031624	Radians Mode

$180^\circ = \pi$ radians

What is Numerical Analysis?

Numerical Analysis is the **study** of **numerical methods** (**algorithms**) that are used to obtain numerical (**approximate**) solutions to mathematical problems.

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When do we Need Numerical Methods?

1. No analytical solution exists.

2. Obtaining an analytical solution is difficult or not practical.

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Solution Types

Analytical Solution (Exact / Accurate Solution)

Some simple equations can be solved analytically.

EXAMPLE

Find the roots of the following equation:

$$x^2 + 4x + 3 = 0$$

$$x = \frac{-4 \quad \sqrt{4^2 - 4 \times 1 \times 3}}{2}$$

$$x = -1 \; and \; x = -3$$

Solution Types

Numerical Solution (Approximate Solution)

Many other equations have no analytical solution.

EXAMPLE

$$x^9 - 4x^2 + 5 = 0$$

$$x - e^{-x} = 0$$

$$x = e^x - x - 2 = 0$$

$$\frac{\sin(e^x) + \sqrt{\cos(\log x^{13})}}{log(\sin(log(x^{77}\sqrt{x^4})))} = 0$$

Solution Types

Numerical Solution (Approximate Solution)

EXAMPLE

Based on the following data, approximate the value of f(25), and f'(30).

x	f(x)
10.1	-0.6250
22.2	-0.2073
32.3	0.7733
42.4	-0.9999

Normalized Decimal Form

A number is **normalized** when it is written in scientific notation with one non-zero decimal digit before the decimal point. Thus, a real number, when written out in normalized scientific notation, is as follows:

$$\pm \; d_0.d_1d_2d_3 \ldots\ldots \times 10^n$$

where **n** is an integer, $d_0d_1d_2d_3$ are the digits of the number in base 10, and d_0 is not zero.

EXAMPLE

$$1.02364 \times 10^3$$

Normalized Decimal Form

EXAMPLE

Are the following numbers normalized?

1.0×10^{-9}

0.1×10^{-7}

10.0×10^{-10}

5.08×10^{3}

5.67×10^{-4}

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Normalized Decimal Form

EXAMPLE

Represent the following numbers in NDF.

0.000567

714.06

0.005008

Idea

move the decimal point to the left or right until there is only one non-zero digit to the left of the point (.) and then compensate for it in the exponent.

Error Measures

Compute $\sqrt{2}$ 1.414213562

Assume a student approximate $\sqrt{2}$ by 1.4

Compute the error between the exact value and approximation one.

Error Measures

Suppose that P^* is an approximation to P, then there are three measures to compute the error:

Absolute Error (AE) =	$\lvert P - P^* \rvert$
Relative Error (RE) =	$\lvert P - P^* \rvert / \lvert P \rvert;\ P \neq 0$
Relative Percentage Error (RPE) =	$RE * 100\%;\ P \neq 0$

Example

Example

Determine the absolute, relative and relative percentage errors when approximating P by P^* when $P = 3$ and $P^* = 3.1$.

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Example

Example

Determine the absolute, relative and relative percentage errors when approximating P by P^* when $P = 0.37856 \times 10^1$ and $P^* = 0.379 \times 10^1$

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Example

Example

Determine the absolute, relative and relative percentage errors when approximating P by P^* when $P = 0.37856 \times 10^{-3}$ and $P^* = 0.379 \times 10^{-3}$

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Example

Example

Determine the absolute, relative and relative percentage errors when approximating P by P^* in each of the following cases:

Case	P	P^*
a	2.1666	2.1667
b	0.0004	0.0003
c	10000.0001	10000

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Example - Solution

Case	AE	RE	RPE
a			
b			
c			

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Rounding and Chopping

If y is a number presented as :

$$y = \underbrace{i_1, i_2, i_3, \ldots}_{\text{Integer Part}} \underbrace{\quad . \quad}_{\text{Decimal Point}} \underbrace{d_1, d_2, d_3, \ldots, d_k, d_{k+1}, d_{k+2}, \ldots}_{\text{Fractional Part/Decimal Part/Mantissa}}$$

Then the floating-point form of y, denoted fl(y), is obtained by terminating the mantissa of y at k decimal digits. There are two common ways of performing this termination:

- Chopping:
  Simply chop off (ignore) the digits $d_{k+1} d_{k+2} \ldots$

- Rounding:
  Add 1 to d_k if $d_{k+1} \geq 5$ else do chopping.

Rounding and Chopping

0.32462156

3-digit chopping

0.324

3-digit rounding

0.325

Example

Example

Determine the 5-digit (a) chopping and (b) rounding values of the irrational number (π).

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Finite-Digit Arithmetic

Example

Use 5-digit chopping arithmetic to find the value of (z). Compute the relative percentage error.

$$z = \frac{1}{3} + \frac{5}{7}$$

RPE =

Finite-Digit Arithmetic

Example

Use 3-digit rounding and 3-digit chopping to evaluate:

$$a = \frac{\frac{13}{14} - \frac{6}{7}}{2e - 5.4} = 1.95354$$

Compute absolute and relative errors.

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Example Solution

		Rounding	Chopping

Example Solution

Errors When Rounding

P =	
P* =	

AE =	
RE =	
RPE =	

Example Solution

Errors When Chopping

P =	
P* =	

AE =	
RE =	
RPE =	

Example

Example

Suppose that $x = \frac{5}{7}$ and $y = \frac{1}{3}$. Use five-digit chopping for calculating the following expressions. All your answers should be in a normalized normal form. In each case compute the absolute and relative errors.

1. $x + y$
2. $x - y$
3. $x \times y$
4. $x \div y$

Example Solution

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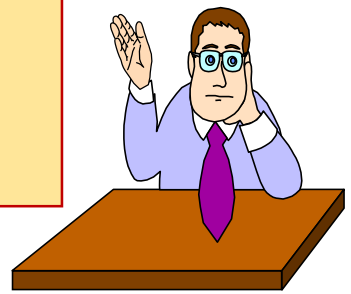
Exercise

Use 3-digit rounding to compute the value of x.

$x = -4 \sin 30 + 5e^2 - 3/17$



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Exercise

Example

Perform the following calculations:

1. exactly.
2. using three-digit chopping arithmetic.
3. using three-digit rounding arithmetic.
4. Compute the relative errors in parts (ii) and (iii).

For:

1. $\frac{4}{5} + \frac{1}{3}$
2. $\frac{4}{5} \times \frac{1}{3}$
3. $(\frac{1}{3} - \frac{3}{11}) + \frac{3}{20}$
4. $(\frac{1}{3} + \frac{3}{11}) - \frac{3}{20}$

Check - Chapter Learning Outcomes

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End of Chapter

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